Encrypted gradient-boosted trees: what the node comparison costs

1 Approximate comparison

41.4

of 1,000 decisions differ
at degree 9 (simulated)

19.6% of the plaintext positives lost;
the ranking is largely kept;
degree 63 did not remove it

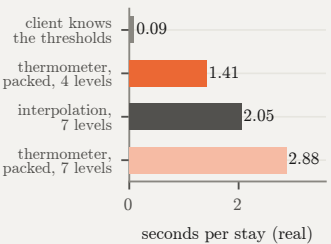
2 Exact comparison, quantized inputs

0

differing decisions at 5, 8 and 16 bins
(simulated, thresholds between margins)

cost: AUROC lower by 0.0095 to 0.0043;
exact for the quantized model only;
the first run had ties (rounding)

3 Thresholds hidden, honest client



on one chain the three circuits trade
time against upload

Proxy cohort (MIMIC-IV-ED-derived), 5 training seeds; no clinical performance is claimed.